

KOUSHIK ANUMALLA

Data Analyst | 3+ YOE | Python · AWS · SQL · Data Analysis

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EDUCATION

M.S. in Computer Science (Machine Learning)

University of Illinois

Aug 2023 - May 2025

GPA: 3.95/4.0

WORK EXPERIENCE

Database Intern | Illinois Department of Human Services | Chicago, IL | Sep 2024 - May 2025

Tech Stack: SQL, Python, SQL Server, Excel, PySpark, Data Analytics, Power BI

- Extracted and analyzed Medicaid and treatment datasets using SQL and Python to support regulatory compliance and public health reporting.
- Optimized SQL queries and developed dashboards in Power BI and Excel to monitor data quality and program effectiveness.
- Collaborated with cross-functional teams to improve data accuracy and streamline reporting workflows.

System Engineer | Tata Consultancy Services (TCS) | India | Aug 2021 – Jul 2023

Client : Nationwide Insurance

Tech Stack: Python, SQL, PySpark, JIRA, ServiceNow, Agile, Data Modeling

- Executed statistical analysis, data modeling, and data mining techniques on 10M+ insurance records to drive forecasting accuracy by 18%.
- Utilized Python, R, and SQL for deep data analysis, delivering insights that improved operational efficiency by 22%.
- Designed dynamic dashboards using Power BI, enabling non-technical stakeholders to act on real-time insights.
- Interpreted complex data sets to guide cost-saving strategies, reducing overhead by 12%.
- Solved data quality issues with meticulous validation, enhancing report reliability.
- Juggled 5+ projects using Jira and Agile, ensuring on-time delivery.
- Collaborated cross-functionally and continuously adopted new data tools to elevate analysis impact.

Summer Intern | Microsoft Technical Associate Program (via Codemania) | India | Jun 2019 - Aug 2019

Tech Stack: R, Python Libraries, Machine Learning, Data Analytics

- Improved topic modeling accuracy by 21% through text data cleansing and feature engineering on 50,000+ airline reviews using Python NLP libraries.
- Automated data workflows with clear documentation ensuring reproducibility and data quality for downstream analysis.

PROJECTS

Bank Marketing Data | KNN Classification , R | [GitHub](#)

- Conducted EDA, handled missing data, applied one-hot encoding, used cross-validated hyperparameter tuning, addressed class imbalance with SMOTE, and benchmarked KNN, logistic regression, and random forest via ROC-AUC.

Machine Learning Projects | Supervised, Unsupervised Algorithms, R, Python | [GitHub](#)

- Developed 7+ models (classification, regression, clustering) across real-world datasets including healthcare, finance, and customer feedback.

Weather Data ETL Pipeline | Apache Airflow, PostgreSQL, Open-Meteo API | [GitHub](#)

- Created an automated pipeline extracting weather data, transforming it into normalized tables for analytics.

Real-Time Stock Market Data Pipeline | Apache Kafka, AWS, Python | [GitHub](#)

- Built real-time ingestion and transformation workflows using streaming architecture and cloud-native tools.

SKILLS

Technical Skills: Python, SQL, MySQL, SQL Server, PostgreSQL, ETL, Data Modeling, PySpark, Informatica, Power BI, Git, Jira, ServiceNow, Data Warehousing, Data Quality, Data Visualization, SDLC, Agile

Cloud Platforms: AWS (S3, Glue, Redshift, EMR, Lambda)

ML & GenAI: Data Analysis, Pandas, Scikit-learn, Prompt Engineering

CERTIFICATIONS

- AWS Certified Cloud Practitioner – Issued Mar 2025 [[Link](#)]
- AWS Certified AI Practitioner – Issued Apr 2025 [[Link](#)]

PUBLICATIONS

- Enhancing **CNN Model** for Automatic Detection of Breast Cancer Using Histopathology Images.